DATA ITEM DESCRIPTION

Title: HEMP PROTECTION SUBSYSTEM (HPS) PERFORMANCE IMMUNITY TEST PLAN

Number: DI-EMCS-81853A
AMSC Number: 10414
DTIC Applicable: No
Preparing Activity: DS

Approval Date: 20230320
Limitation: N/A
GIDEP Applicable: No
Project Number: EMCS-2023-009

Use/Relationship:

The HEMP Protection Subsystem (HPS) Performance Immunity Test Plan applies to military standard MIL-STD-3023A, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Military Aircraft*. DI-EMCS-81853, *Test Plan for Mission Critical Subsystem Immunity Test*, is hereby cancelled and superseded by DI-EMCS-81853A. This DID contains the format, content, and intended-use information for the data deliverable resulting from the immunity test tasks described in MIL-STD-3023A, Appendix C, sections C.4.4 through C.4.7 and C.5.3, as applicable.

The HEMP Protection Subsystem (HPS) Performance Immunity Test Plan shall prescribe the detailed procedures for planning, preparation, implementation, and reporting of MCS immunity testing as specified by MIL-STD-461 or equivalent. The information provided in this HEMP Protection Subsystem (HPS) Performance Immunity Test Plan will constitute adequate proof, when combined with data analysis, that the MCS immunity testing requirements in MIL-STD-3023A, Appendix C, have been achieved. The objective of MCS immunity testing is to provide the immunity data in terms of its norm attributes for assessing the design margin requirements in MIL-STD-3023A. (Definition of norm attributes is contained in MIL-STD-3023A, Appendix C.)

The HEMP Protection Subsystem (HPS) Performance Immunity Test Plan provides the procedures used by the government to obtain and evaluate HEMP protection performance data on MCS to be installed in aircraft.

This DID is related to DI-EMCS-81850A, DI-EMCS-81851A, and DI-EMCS-81854A.

This DID supersedes DI-EMCS-81853.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and any applicable amendments, notices, and revisions, shall be as specified in the contract. The HEMP Protection Subsystem (HPS) Performance Immunity Test Plan classification shall be determined using the DTRA Security Classification Guide for DoD Electromagnetic Pulse (EMP) Programs and Activities (U) available by mail request to ATTN: RD-NTSA, Defense Threat Reduction Agency, 8725 John J. Kingman Road, MSC 6201 Fort Belvoir, Virginia 22060-6201, and any relevant system specific classification guides.

2. Format. The HEMP Protection Subsystem (HPS) Performance Immunity Test Plan shall be in contractor format.

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3. Content. The HEMP Protection Subsystem (HPS) Performance Immunity Test Plan shall describe the overall test methods and procedures for the MCS immunity test for each requirement specified in the contract for the aircraft being developed.

3.1 Summary information. This report shall summarize the following:

3.1.1 Introduction.

a. MCS description including any pertinent test issues.

b. Statement of any assumptions and limitations associated with MCS immunity tests.

3.1.2 A general description of the test methods and procedures shall be provided for immunity testing each MCS and performance requirement area listed in section 3.2 below.

3.2 Detailed information. For each MCS being immunity tested, a test plan shall be prepared. The test plan shall contain comprehensive, MCS-specific detailed test procedures. The document shall contain the following information:

a. A statement of test objectives and criteria to be met to achieve these objectives.

b. MCS identification and description including drawings showing locations of all points of entry (POEs), external cable interconnections, and a description of installed HEMP or other electromagnetic protection measures (if any).

c. When and where test will be conducted.

d. Who will conduct the test.

e. Government witness(es) to the testing, if any.

f. A discussion of any planned deviations from the test plan requirements of MIL-STD-3023A, Appendix C.

g. Requirements for test results including all raw and processed test oscillographic or digital waveform data in engineering units. The electronic data format used to capture the test data will be described in the test plan.

h. Data acquisition equipment requirements (including manufacturer, model, characteristics, detailed calibration procedures and calibration traceability documentation).

i. Reference to detailed step-by-step test procedures to be used to include procedures for determining Hardness Protection Subsystem performance test immunity norms and the approach to calculating norms-based design margins as required by MIL-STD-3023A.

j. Any deviations from the requirements of MIL-STD-3023A.

k. Data management requirements (including data quality control procedures, data acceptability criteria, annotation and preservation of data records, and pass/fail criteria).

l. Safety, including electromagnetic radiation and electrical shock hazards.

m. Security procedures.

n. Schedule.

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End of DI-EMCS-81853A.

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